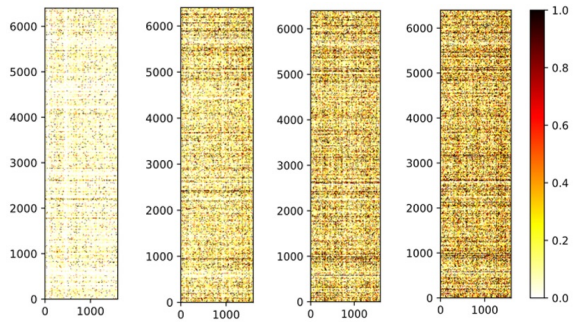


(a) 1 edit (b) 5 edits (c) 10 edits (d) 15 edits

Figure 5: Visualization of the distinction between the final edited weight and the original unedited weight via weight change $|\Delta W|$ as the number of edits increases.

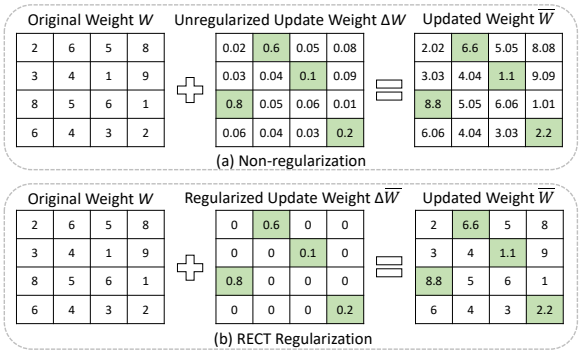


Figure 6: Comparison of (a) non-regularization and (b) the proposed RECT regularization. Elements in green denote the top-$k\%$ that change the most according to δ and are considered as the principle editing information,which should keep their original values. k = 25 in this figure for illustration.

across multiple edits can amplify changes to the original weights.

Visualization The distinction between the final edited weight and the original unedited weight is illustrated by visualizing the weight change $|\Delta W|$ as shown in Figure 5. It reveals the consistent findings that the update weight ΔW might be quite sparse, while the accumulation across multiple edits can amplify changes to the original weights.

5 RECT: RElative Change in weighT

5.1 Approach

We have analyzed the causes of side effects in Section 4.4 that model editing changes the original model weights too much, resulting in *overfitting* to the editing facts. This type of editing overfitting occurs when a model learns to fit the new editing data too closely, capturing noise and outliers in the data rather than the underlying patterns. Furthermore, the gradual buildup of editing-induced overfitting across sequential edits can severely impair the general abilities of LLMs. Consequently, while such models may exhibit proficiency in the new editing facts, they often struggle to generalize effectively across a spectrum of downstream tasks. This phenomenon underscores the importance of mitigating overfitting during the editing process to ensure both the improvement of model factuality and the maintenance of their general abilities.

To this end, this paper designs a regularization method named **RE**lative **C**hange in weigh**T** (RECT) to prevent editing overfitting. Figure 6 illustrates the overview of this regularization method. Typically, one editing operation is to add an edit update weight ΔW to the original weight W to derive the updated weight $\overline{W}$, where ΔW is calculated aiming to insert a batch of N new editing facts $\{(s, r, o^*)_i\}_{i=1}^{N}$ ($N = 1$ for a single editing fact). Formally, we have:

$$\overline{W} = W + \Delta W, \tag{1}$$

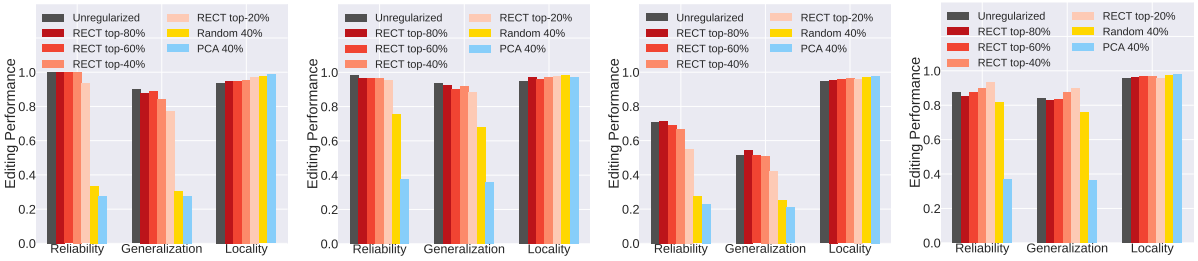
$$\Delta W = f(\{(s, r, o^*)_i\}_{i=1}^{N}), \tag{2}$$

where function f denotes the calculation method of update weight ΔW for different editing methods, e.g., ROME (Meng et al., 2022).

Here, we define the absolute value of the relative change in weight $\delta = |\frac{\Delta W}{W}|$ to characterize the degree of change of each element in the update weight ΔW. To some extent, δ can be used to indicate the importance of each element in ΔW when inserting the new editing facts. On the one hand, a portion of the elements in the update weight $\overline{W}$ are assumed to constitute the core components of the new editing facts. On the other hand, the remaining elements are assumed minor contributions to editing. Specifically, the top-$k\%$ elements in ΔW that change the most according to δ are considered as the principal editing information and keep their original values. While for the remaining elements in ΔW, they are treated as minor contributions to editing and set to zero for regularization. Mathematically, we have the *regularized* edit update weight $\Delta\overline{W}$ as:

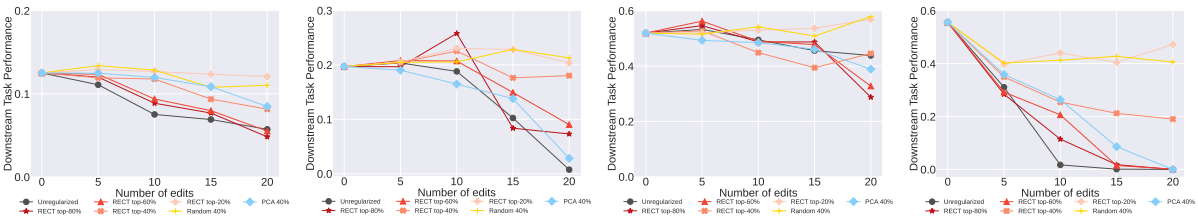
$$\Delta\overline{W}_{ij} = \begin{cases} \Delta W_{ij} & \text{if } \delta_{ij} \text{ in the top-}k\%, \\ 0 & \text{else.} \end{cases} \tag{3}$$

Finally, the regularized edit update weight $\Delta\overline{W}$ is added to the original weight to derive the regularized updated weight. Essentially, RECT functions



(a) ROME on GPT-2 XL (b) ROME on LLaMA-1 (7B) (c) MEMIT on GPT-2 XL (d) MEMIT on LLaMA-1 (7B)

Figure 7: Comparison of introducing various regularization methods and how the *editing performance* change with respect to different top-$k\%$ for RECT.



(a) Summarization (b) Open-domain QA (c) Closed-domain QA (d) Sentiment Analysis

Figure 8: Comparison of introducing various regularization methods and how the *downstream task performance* change with respect to different top-$k\%$ for RECT.

to deter the implementation of excessively intricate editing updates that have a higher propensity to result in overfitting. By imposing constraints on the complexity of editing updates, it serves as a safeguard against the model's inclination to adapt too closely to the editing data, thus promoting more generalizable and reliable model performance.

5.2 Regularization Baselines

To demonstrate the effectiveness and efficiency of the proposed method RECT, we compared it with the following baselines, including: **Unregularized** keeps the full elements of the edit update weight ΔW. **Random $k\%$** selects the random $k\%$ elements of ΔW. **PCA $k\%$** compresses the most important editing information in ΔW into $k\%$ elements via principal component analysis (PCA), and sets the remaining elements to zero.

5.3 Results of RECT

The effectiveness of a regularization method should be illustrated from two perspectives. First, regularizing the edit update weight should not harm its editing performance, i.e., edited models should still remember the new editing facts and generalize to related facts. Second, the regularized edited models should be able to preserve the general abilities compared with unregularized ones.

Editing Performance Figure 7 presents the the results of regularizing ROME or MEMIT on GPT-2 XL or LLaMA-1 (7B). Readers can refer to Appendix D.2 for the results on LLaMA-2 (7B). From these results we can have the following findings. First, compared with unregularized ΔW, RECT that keeps the original values of an appropriate amount of top-40% elements in ΔW and sets the remaining elements to zero can help maintain over 94% majority of reliability and generalization, and even improve locality. It is natural that reliability and generalization slightly drop when setting partial elements to zero since partial editing information is removed. The reason why locality is improved is probably because those elements with low δ corresponding to some noise and outliers in the editing data are removed to prevent from editing overfitting, so the edited models are more robust. However, setting excessive elements in ΔW to zero, e.g., RECT top-20%, might hurt the editing performance as partial important editing information is accidentally removed. Furthermore, compared with Random 40% and PCA 40%, RECT top-40% achieves the best performance, indicating its effectiveness in selecting the most principal editing information. It is notable that RECT also exhibits advantages in terms of efficiency, since it eliminates the complex calculations required in PCA.

General Downstream Task Performance Figure 8 presents how the downstream task performance change with respect to introducing various regularization methods to edit GPT-2 XL. Readers can refer to Appendix D.2 for the results on more downstream tasks. From these results we can have the following findings. As the proportion of elements in ΔW set to 0 increases, the more editing overfitting is regularized, the smaller the change to the original weight, so the general abilities can be more preserved. Results show that regularized edited models are able to preserve the general abilities compared with unregularized ones in most tasks such as summarization, open- and closed-domain QA. It is worth noting that it still poses a challenge for some tasks such as sentiment analysis, and remains unclear whether it works for larger number of edits, which will be left to future work.

6 Conclusion

Model updating technology has been catalyzing the continuous iteration of advanced and trustworthy LLMs. This paper studies model editing and for the first time raises concerns whether model editing has any side effects on the general abilities of LLMs. The systematical evaluation reveals that current methods unintentionally hurt the general abilities of LLMs no matter in instance- or batch-editing, and single- or sequential-editing. Our analysis of the causes reveals that model editing results in overfitting to the editing facts, and the accumulation of overfitting across multiple edits can amplify the negative impact. The proposed RECT regularization method has been proven to effectively prevent overfitting of new editing facts, thus preserving both the editing and general downstream task performance.

Impact Statements

As LLMs play an increasingly crucial role in various applications, mitigating the hallucinations caused by missing, false or outdated knowledge encapsulated within the parameters is imperative for ensuring the reliability of their outputs. However, the potential trade-off between improving the factuality and degrading the general abilities underscores the need for a balanced approach. Striking the right balance in model editing is crucial to prevent unintended consequences and to preserve the broader abilities of LLMs, contributing to the sustainable advancement of AI technology. This paper highlights the importance of considering not only the immediate gains in factuality but also the long-term impacts on the general performance and applicability of LLMs, encouraging a thoughtful and comprehensive exploration of model editing techniques for responsible AI development. More importantly, this paper calls for more efforts and underscores the collective focus on *strengthening the robustness of LLMs to weight perturbations, developing innovative paradigms for model editing, and designing comprehensive evaluation of model editing*. By doing so, we can collectively advance the continual development of LLMs, paving the way for more reliable applications in real-world scenarios.

Limitations

This paper studies the side effects of editing based on the ZsRE editing dataset, while more complex and diverse side effects are hypothesized to exist and thus need to be explored on more editing datasets in future work. In addition, although sometimes a method of side effect mitigation is effective for a certain number of edits, it remains to be seen whether the method will still be effective for a larger number of edits. It is expected that one editing method outperforms another in terms of the number of edits given the same requirements of maintaining editing performance and general abilities. This paper does not further explore whether the proposed method can still be effective for more edits, which is worth further study. While we primarily propose to mitigate the side effects of model editing from a statistical perspective, the bottleneck of the general abilities of edited models should be analyzed theoretically.

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